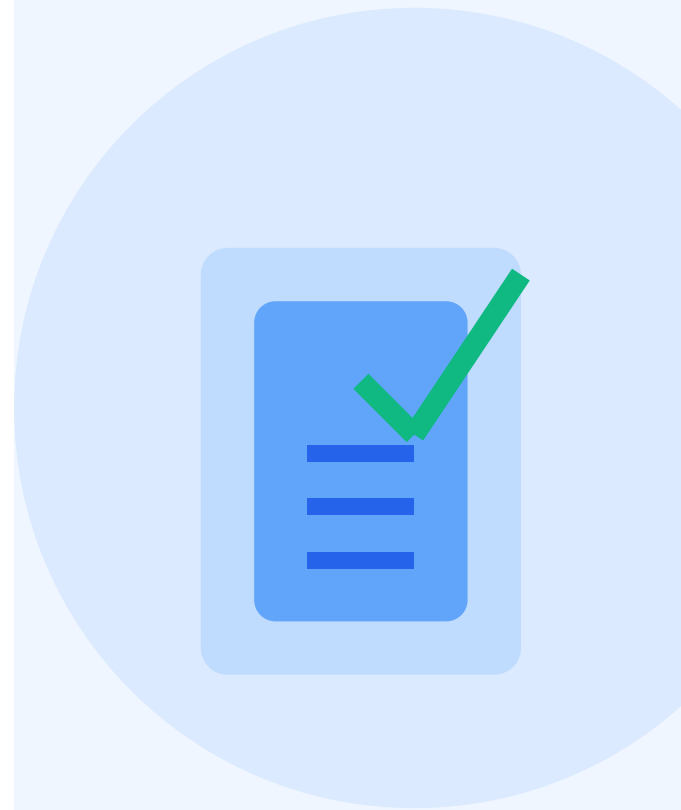


Smart Loan Risk & Approval System

Business Report

End-to-End Machine Learning Application
for Loan Risk Prediction and Approval



Report Summary

Report Type:	Business Report
Generated On:	2026-09-28 21:55
Total Records:	5,000
Total Columns:	21
Target Column:	Loan_Status
Active Model:	Logistic Regression

✓ Data Analysis & Insights

✓ Risk Prediction

✓ Approval Decision

✓ Better Financial Decisions

1. Dataset Overview

Total Records

5,000

Total Columns

21

Numeric Cols

11

Categorical Cols

10

2. Data Quality

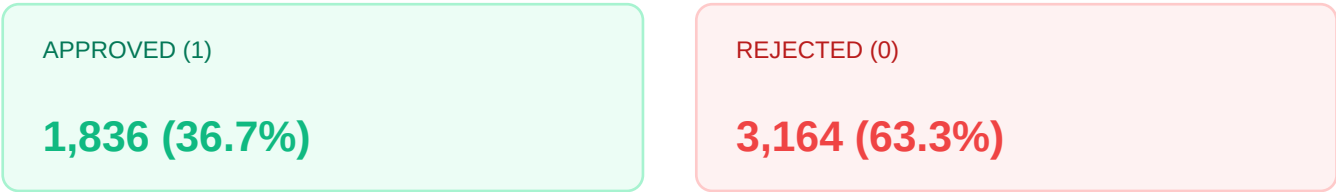
Metric	Count	Percentage
Missing Values	0	0.00%
Duplicate Rows	0	0.00%
Numerical Cols	11	52%
Categorical Cols	10	48%

✓ Dataset is ready for analysis and model training.

3. Missing Values by Column

✓ No missing values detected.
All columns have complete data.

4. Target Analysis (Loan_Status)



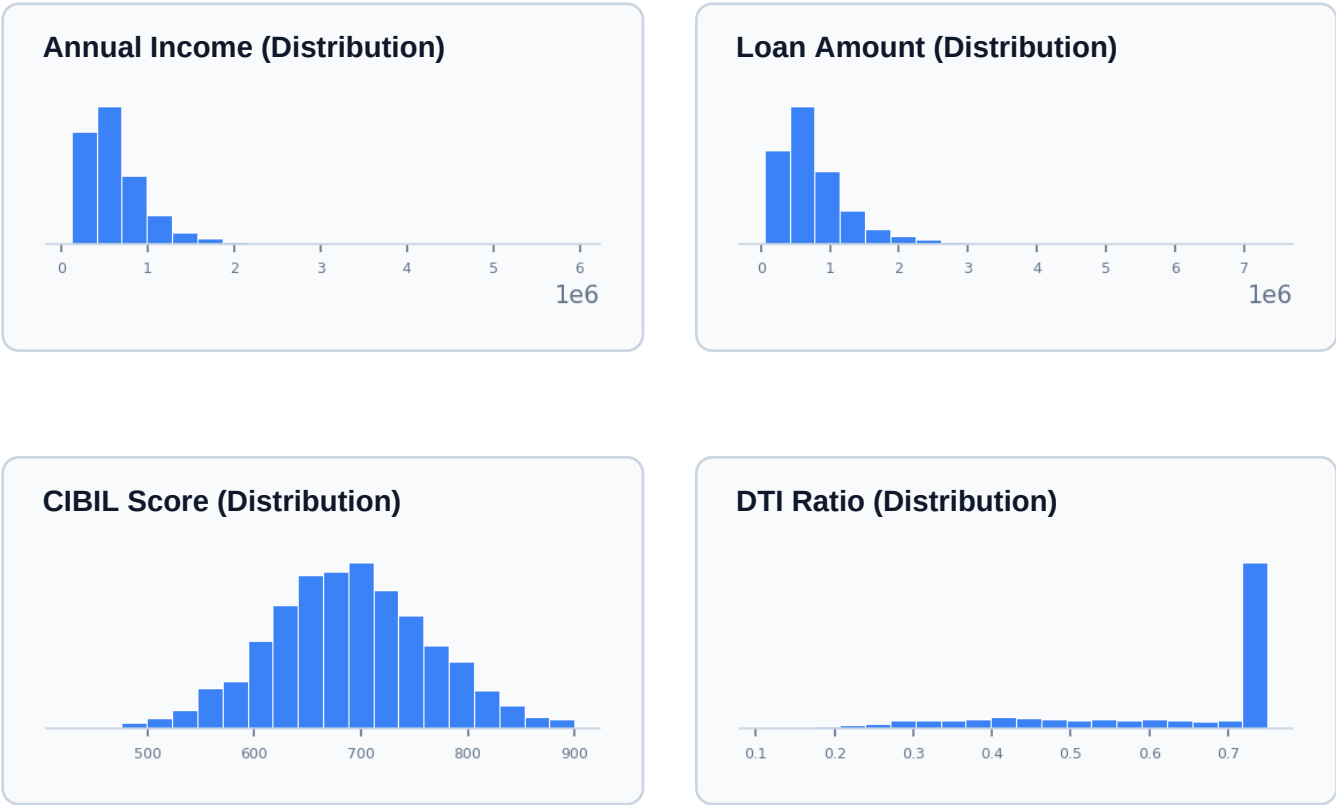
5. Descriptive Statistics

Feature	Mean	Median	Min	Max	Std Dev
Age	42.9	43.0	21.0	65.0	13.0
Employment_Years	15.7	15.1	-0.0	35.0	9.1
Annual_Income	630,699.80	540,000.00	120,000.00	5,942,000.00	375,636.23
Coapplicant_Income	102,987.60	65,000.00	0.00	1,856,000.00	140,083.19
Loan_Amount	792,643.40	638,000.00	51,000.00	7,344,000.00	588,091.34
Loan_Term_Years	8.8	7.0	2.0	20.0	5.3
Cibil_Score	689.4	689.0	429.0	900.0	75.4
Existing_Loans	1.2	1.0	0.0	6.0	1.1
DTI_Ratio	0.6	0.8	0.1	0.8	0.2
Residential_Assets	2,080,798.60	1,369,500.00	66,000.00	80,846,000.00	2,521,706.38
Bank_Assets	478,127.00	343,000.00	22,000.00	7,727,000.00	458,559.07

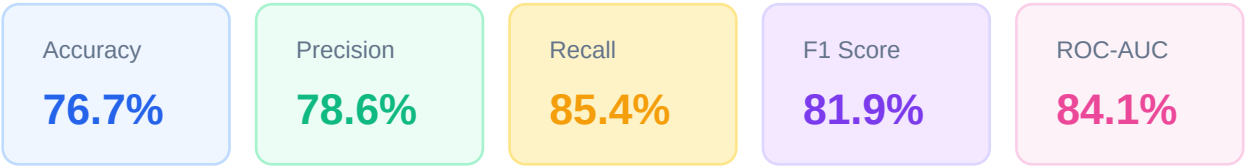
6. Outlier Analysis (IQR Method)

Feature	Outlier Count	Outlier %	Severity
Annual_Income	227	4.5%	Medium
Coapplicant_Income	228	4.6%	Medium
Loan_Amount	282	5.6%	Medium
Cibil_Score	43	0.9%	Low
Residential_Assets	372	7.4%	High
Bank_Assets	309	6.2%	Medium

7. Distribution of Key Features

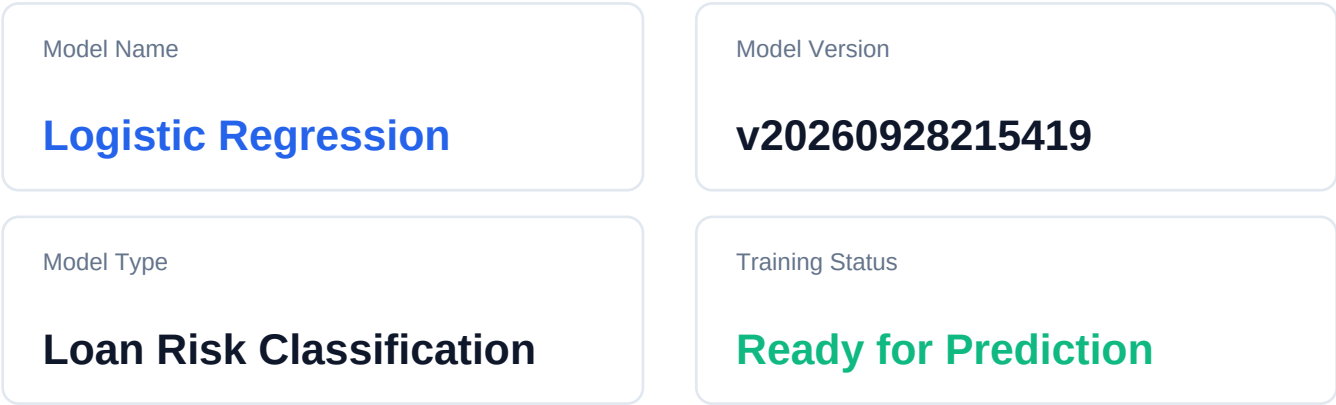


8. Model Performance



Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.767	0.786	0.854	0.819	0.841

9. Active Model Details



10. New Applicant Prediction

Applicant Information		Prediction Result	
Age	24.0	LOW RISK	
Employment Years	4.3	Prediction Confidence: 81.4%	
Annual Income	333000.0	Approval Probability: 81.4%	
Coapplicant Income	0.0	Risk Probability: 18.6%	
Loan Amount	226000.0	Decision: APPROVE	
Loan Term Years	15.0		
Cibil Score	745.0		
Existing Loans	1.0		
DTI Ratio	0.512		
Residential Assets	468000.0		
Bank Assets	127000.0		
Gender	Female		
Married	No		
Dependents	0		

11. Final Summary

The system analyzed 5,000 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset.

Total Records 5,000	Total Columns 21	Selected Model Logistic Regression
Model Accuracy 76.7%	Latest Applicant Result LOW RISK	Approval Probability 81.4%